

Technology Stack

Date	02 July 2026
Team ID	xxxxxxx
Project Name	Credit Card Approval Prediction
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	<i>HTML5, CSS3, Bootstrap, JavaScript</i>	<i>Creates a responsive, user-friendly interface for entering applicant details and displaying prediction results.</i>

2	Backend / Server-Side	<i>Python, Flask</i>	<i>Flask provides a lightweight framework to connect the web interface with the machine learning model and process prediction requests efficiently.</i>
3	Database / Data Storage	<i>CSV Dataset, SQLite (optional)</i>	<i>Stores applicant information and prediction history. CSV is used for model training, while SQLite can be used for application data storage.</i>
4	Cloud / Hosting / Deployment	<i>Render / IBM Cloudlogy]</i>	<i>Deploys the Flask application online, enabling users to access the prediction system from any device with scalability and reliability.</i>
5	Version Control & CI/CD	<i>Git, GitHub</i>	<i>Tracks project changes, enables team collaboration, and manages source code versions efficiently.</i>

6	Third-Party APIs / Other Tools	<i>Scikit-learn, XGBoost, Pandas, NumPy, Matplotlib, Seaborn, Joblib</i>	<i>Provides machine learning algorithms, data preprocessing, visualization, model serialization, and prediction capabilities for the application.</i>